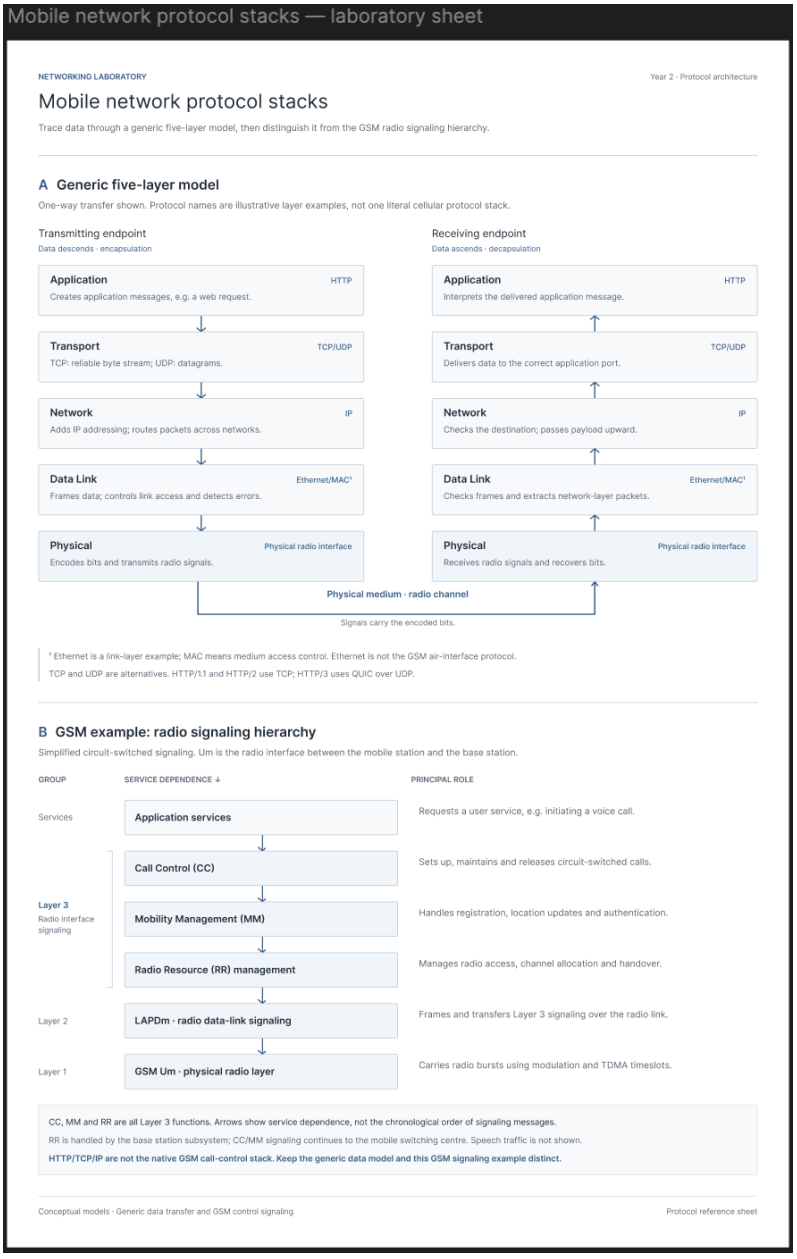
ITA0312 -MOBILE COMPUTING

EXPERIMENT-4

1. Use Figma to create an interactive diagram illustrating the components of a basic cellular system.

Aim:

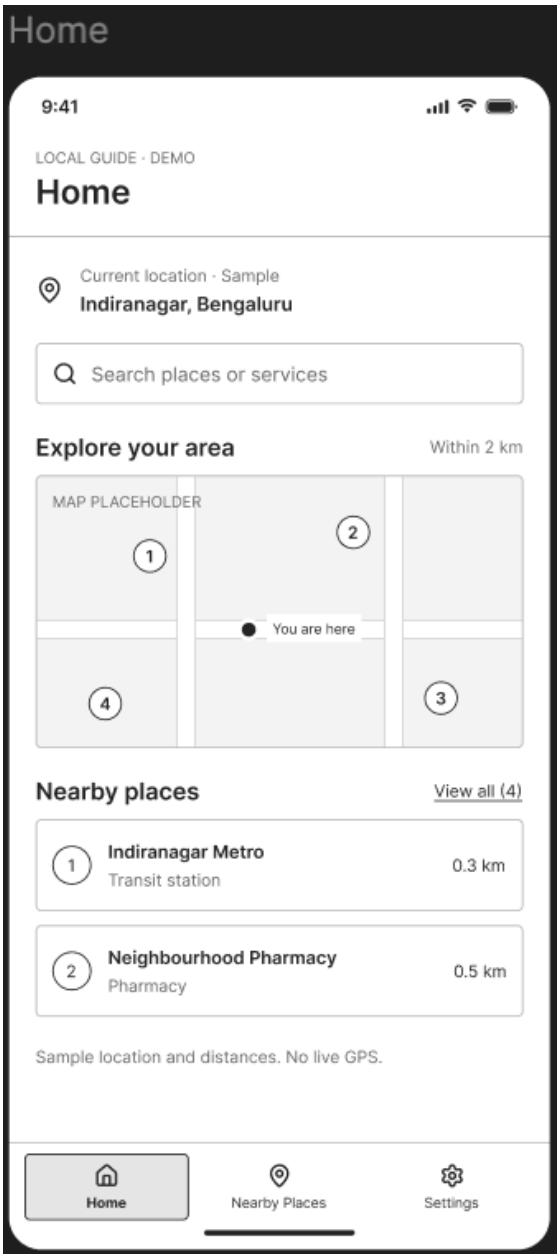
To create a visual representation in Figma, highlighting the key elements and functions of mobile network protocol stacks.

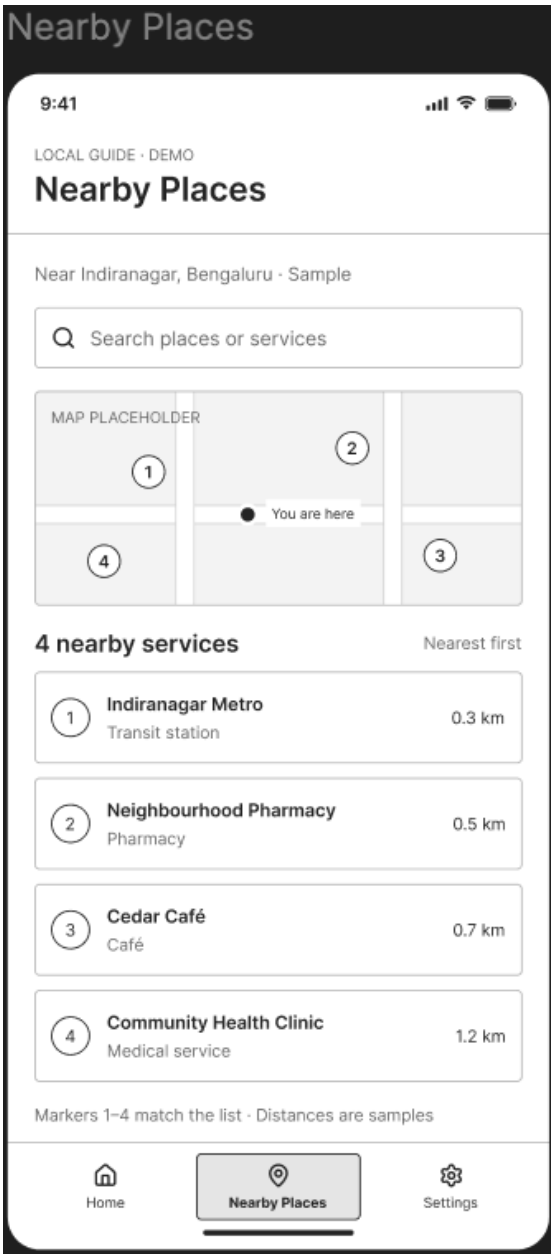


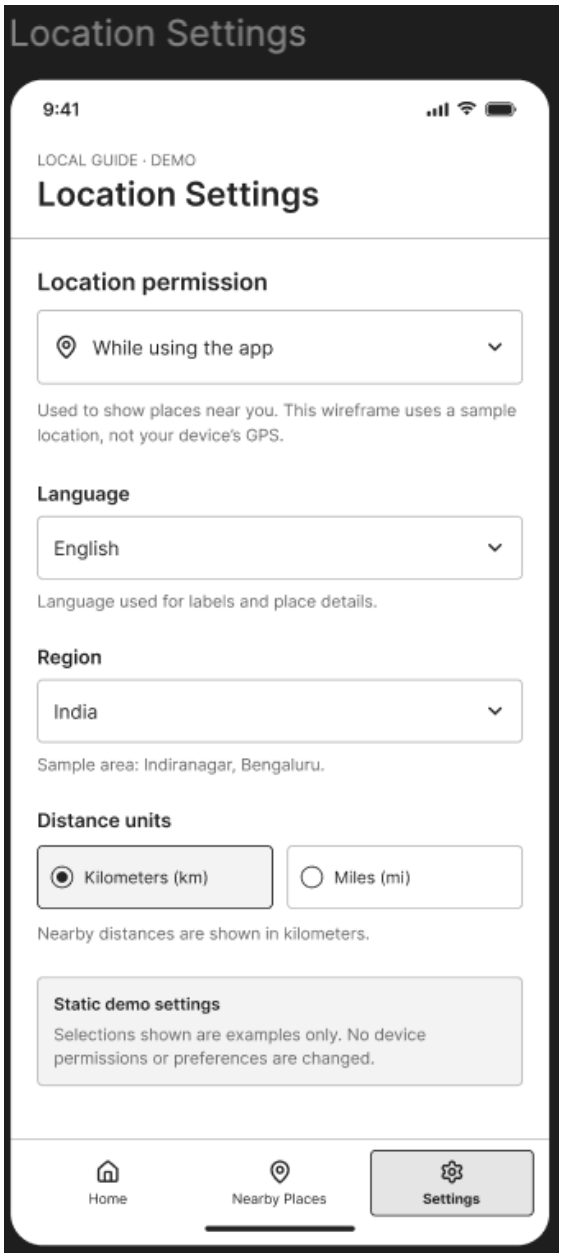
EXPERIMENT-5

Aim:

To design a mobile application wireframe using Figma that incorporates localization features such as location services, language selection, and distance units.



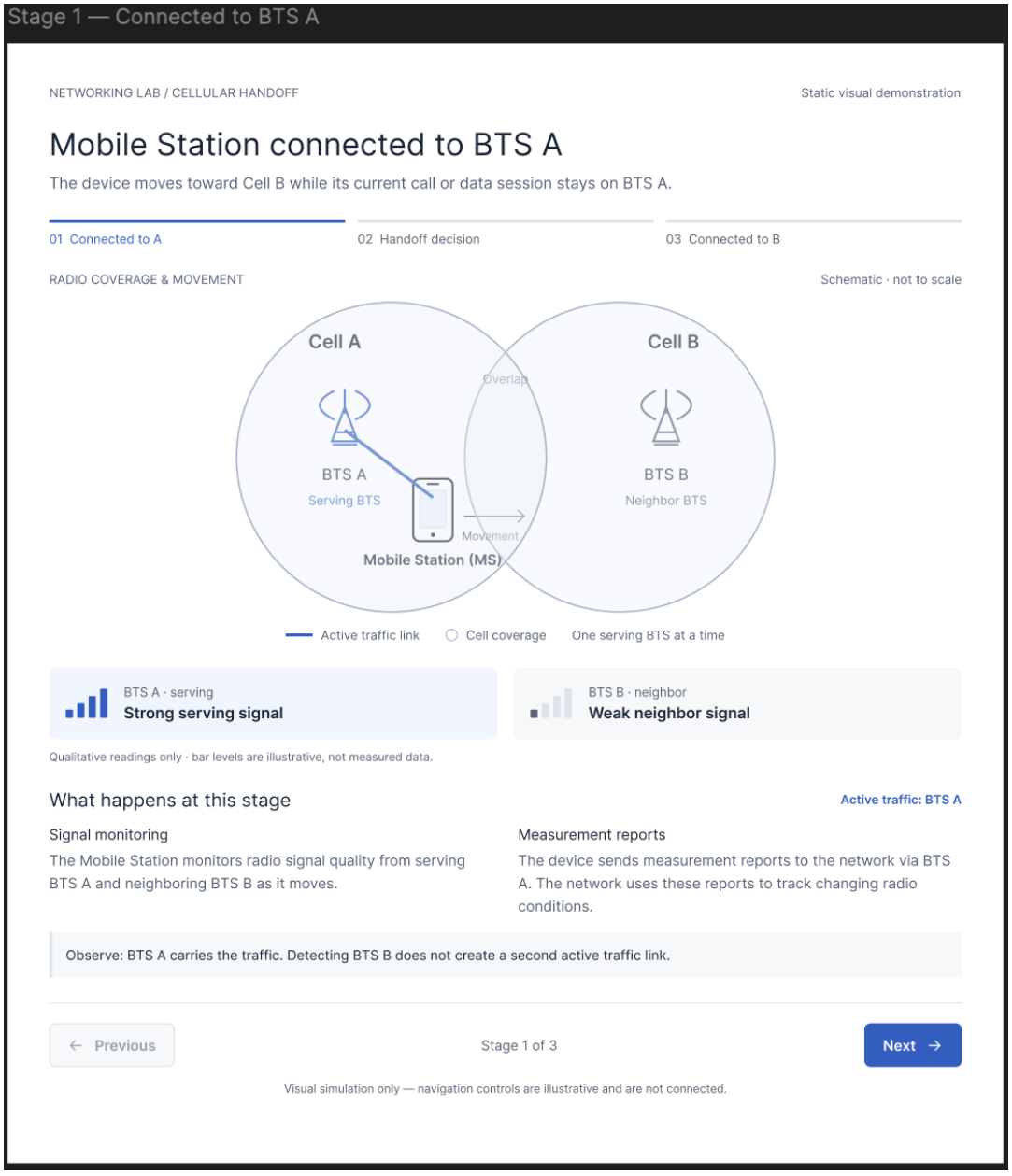



Location Settings
9:41
LOCAL GUIDE · DEMO
Location Settings
Location permission
While using the app
Used to show places near you. This wireframe uses a sample location, not your device's GPS.
Language
English
Language used for labels and place details.
Region
India
Sample area: Indiranagar, Bengaluru.
Distance units
Kilometers (km)
Miles (mi)
Nearby distances are shown in kilometers.
Static demo settings
Selections shown are examples only. No device permissions or preferences are changed.
Home
Nearby Places
Settings

EXPERIMENT-6

Aim:

To simulate and visualize the handoff process in a mobile cellular network using Figma, illustrating the transfer of a mobile station's connection between two cells.



Stage 2 — Handoff decision

NETWORKING LAB / CELLULAR HANDOFF

Static visual demonstration

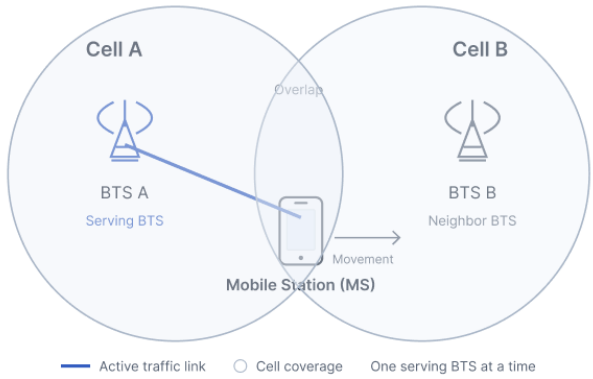
Handoff decision in the overlap

BTS A weakens and BTS B becomes stronger. The network prepares the transfer.

01 Connected to A | 02 Handoff decision | 03 Connected to B

RADIO COVERAGE & MOVEMENT

Schematic · not to scale



BTS A · serving
Weakening serving signal

BTS B · neighbor
Stronger neighbor signal

Qualitative readings only · bar levels are illustrative, not measured data.

What happens at this stage

Active traffic: still BTS A

Network-controlled decision

The network controller evaluates reported radio measurements and handoff criteria, such as signal quality and hysteresis, then selects BTS B.

Target channel allocation

The controller allocates a target traffic channel through BTS B and sends a handoff command to the Mobile Station via BTS A.

> Observe: BTS B is prepared, not yet serving. The handset and the BTS do not independently decide the handoff.

← Previous | Stage 2 of 3 | Next →

Visual simulation only — navigation controls are illustrative and are not connected.

Stage 3 — Connected to BTS B

NETWORKING LAB / CELLULAR HANDOFF

Static visual demonstration

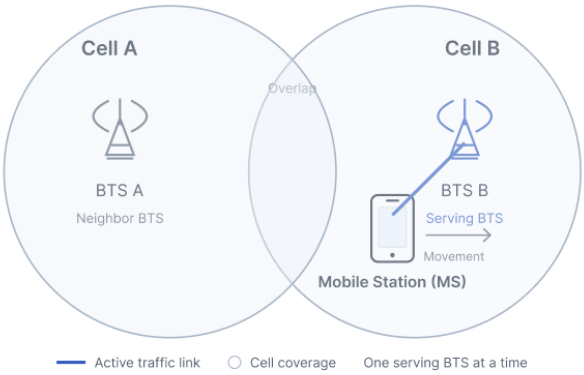
Connection transferred to BTS B

The device is now inside Cell B. BTS B serves the ongoing call or data session.

01 Connected to A | 02 Handoff decision | 03 Connected to B

RADIO COVERAGE & MOVEMENT

Schematic · not to scale



Active traffic link | Cell coverage | One serving BTS at a time

BTS A · neighbor
Weak neighbor signal

BTS B · serving
Strong serving signal

Qualitative readings only · bar levels are illustrative, not measured data.

What happens at this stage

Active traffic: BTS B

Connection transfer

Following the handoff command, the Mobile Station switches to the allocated traffic channel on BTS B and establishes the new radio connection.

Confirmation & completion

The network receives handoff confirmation through BTS B, completes the transfer and releases the old BTS A channel. Communication continues via BTS B.

Observe: only BTS B carries the traffic now. The old A channel is released and is available for reuse.

← Previous | Stage 3 of 3 | Next →

Visual simulation only — navigation controls are illustrative and are not connected.